

PRODUCT & METHODOLOGY REPORT

Churn Intelligence for Telecom Operators

A clustering-informed classification and retention-action system

EXECUTIVE SUMMARY

Turning a low-recall churn model into a retention engine

Subscriber churn is the single largest controllable drain on a telecom operator's revenue. This report documents an end-to-end product that does more than flag who might leave: it groups customers into behavioural **archetypes**, predicts churn risk with a recall-first classifier, explains *why* each at-risk customer is likely to leave using **SHAP**, and then drafts a concrete, tailored retention action for every flagged customer using a large language model (**gpt-oss-120b**) served through the **Groq API**.

0.928

Recall on churners (LightGBM)

26.5%

Base churn rate in data

4

Behaviour archetypes found

606

At-risk customers actioned

The core narrative is a fix. A single classifier tuned only for headline accuracy quietly let too many churners slip through — a low-recall failure that is expensive precisely because the customers it misses are the ones worth saving. The remedy in this project is not a bigger model but a better pipeline: cluster the customer base into archetypes, carry that segment context into a recall-first classification stage, and use the resulting high-confidence risk pool to power targeted, explainable retention outreach. The result is a system that catches roughly nine of every ten churners and hands the retention team a ready-to-use reason and remedy for each one.

What ships in the box

A merged, leakage-audited feature set · four validated customer archetypes (UMAP + HDBSCAN) · a five-model classification bake-off with a recall-first operating point · per-customer SHAP driver attribution · and an LLM layer (gpt-oss-120b on Groq) that writes both per-customer briefs and reusable per-archetype retention playbooks — exported to a single spreadsheet the retention team can act on.

SECTION 1

The Problem: Why We Are Building This

For a telecom operator, acquiring a subscriber costs many times more than keeping one. Yet churn in the industry runs high and is often noticed only after the customer has already gone — the cancellation call, the ported number, the final invoice. By then the intervention window has closed. The commercial problem is therefore not "who left?" but "who is *about* to leave, why, and what single action would keep them?" — answered early enough and specifically enough that a retention team can act on it at scale.

Most churn projects stop at a risk score. A number between 0 and 1 tells an operator that a customer is risky but not *why*, and certainly not *what to do*. A retention agent handed a list of 5,000 "risky" customers with no reasons and no recommended offers cannot act differently for each one, so the list goes unused. This product is built to close that gap end to end: from raw subscriber tables to a per-customer reason and a per-customer recommended move.

Who this is for — the target segment

The customer for this product is the **telecom operator** — mobile, broadband, and fixed-line providers — and specifically the teams inside them that own subscriber revenue: retention and customer-success groups, CRM and marketing operations, and the revenue-assurance analysts who quantify churn's cost. These organisations already hold exactly the data the pipeline consumes: demographics, service subscriptions, usage, billing, and geography. What they typically lack is a system that converts those tables into a prioritised, explained, and *actionable* list. The dataset used throughout this report is IBM's public Telco customer-churn set (7,043 customers, five source tables), which mirrors the shape of a real operator's warehouse and makes the workflow directly portable.

The cost of the status quo

At a 26.5% base churn rate — the rate in this dataset — more than one subscriber in four is lost over the observed period. Every churner the model fails to flag is a save the operator never attempts. That is **why recall** — the share of true churners the system actually catches — is the metric this product optimises first, ahead of a prettier accuracy figure. A model that misses churners is worse than useless: it creates false confidence while revenue walks out the door.

SECTION 2

Where a Single Classifier Falls Short

The intuitive first attempt at churn prediction is a single classifier: pour every feature into one model, train it, and threshold its output at the default 0.5. This is where most projects begin — and it is where this one ran into a wall. The churn is imbalanced (only ~27% positives), and at the default cut the model quietly optimises for the majority — the customers who stay — while letting a substantial share of genuine churners fall below the threshold. The headline number looks fine; the **recall on the class that matters is poor**. In business terms, the model misses the very customers it exists to catch.

Hence, we first performed clustering to get the customer behavioural archetypes using **UMAP+HDBSCAN**, and then trained classifiers based on these archetypes along with the rest of the features.

The project was built around a recall floor. Rather than accept whatever recall the default threshold happened to give, the operating threshold is chosen explicitly — always out-of-fold, never on the test set — to hold recall at or above 0.80 while keeping precision at a level a retention team can staff. That single discipline is what turns a good-looking-but-leaky classifier into one that actually catches churners.

SECTION 3

The Pipeline: Cluster → Classify → Explain → Act

The product is a four-stage pipeline. Each stage feeds the next, and clustering — rather than being a dead-end feature — threads through all of them as the source of behavioural context.

Stage	What happens	Output it hands on
1 · Data foundation	Merge five IBM source tables; audit and drop leakage, duplicates, and constants.	A clean 31-feature modelling set + a quarantined reference file.
2 · Archetypes	UMAP + HDBSCAN on a churn-relevant feature subspace to find customer behaviour segments.	4 stable archetypes, each with a distinct churn profile.
3 · Classification	Five-model bake-off predicting per-customer churn probability, tuned recall-first.	A calibrated risk score + a high-confidence at-risk pool.
4 · Explain & act	SHAP attributes each risk to its drivers; gpt-oss-120b (Groq) drafts the retention move.	Per-customer brief + per-archetype playbook, exported to Excel.

Figure 1. The four-stage pipeline. Clustering (stage 2) supplies the behavioural context consumed by the explanation and action stages.

3.1 Building an honest feature set

Before any modelling, five IBM tables — demographics, location, population, services, and status — are merged into one row per customer. The critical work here is removing **leakage**: fields that were recorded with or after the outcome and would let the model "cheat." For example, IBM's *Satisfaction Score* scores 0.95 AUC on its own and *Churn Score* 0.94 — versus ~0.86 for a fully tuned model on real features. Those are the outcome wearing a disguise; you will not have next quarter's satisfaction survey when scoring next quarter's customers, so they are dropped. Exact duplicates (e.g. *Under 30* is just *Age < 30*), constants, and identifiers are removed too — each one proven redundant against the data rather than dropped on faith. The result is a 31-feature set with zero leakage and a 26.5% churn rate.

3.2 Reading customer behaviour archetypes (clustering)

Customers are grouped with **UMAP** (dimensionality reduction) followed by **HDBSCAN** (density clustering). Crucially, the clustering space is not the full feature set: columns earn their place by their measured association with churn (point-biserial correlation for numeric features, Cramér's V for categorical), computed on the training set only. This keeps the archetypes aligned to retention-relevant behaviour. A hyperparameter search over 40 trials selected a four-cluster solution with **0% noise** and a guarded silhouette of 0.84; a bootstrap stability check (re-clustering 80% resamples from scratch) returned a mean ARI of **0.933** — the same customers cluster together run after run.

Archetype	Share	Churn rate	Defining profile
0 · Low-value, no-internet	21.8%	7.5%	Low bill (~\$21), no internet, credit-card payer — loyal, low risk
1 · DSL mainstream	39.2%	18.5%	DSL, married, mid bill (~\$63), has protection — the stable core
2 · Fibre high-value family	22.7%	38.5%	Fibre, high bill (~\$93), few dependents — competitive, at risk
3 · Fibre solo, no referrals	16.3%	54.5%	Fibre, unmarried, zero referrals — isolated, highest churn

Table 1. The four behaviour archetypes. Churn rate rises from 7.5% to 54.5% across the segments — evidence the clusters capture real, retention-relevant structure (overall base rate 26.5%).

These archetypes are the product's human layer. A raw probability of 0.82 means little to a retention agent; "an *Archetype 3* customer — fibre, solo, no referrals, in a segment that churns at 54%" is a story they can act on. That peer context is exactly what the action stage consumes.

3.3 Classifying churn risk — the four-model bake-off

Five models predict each customer's churn probability: Logistic Regression, Decision Tree, Random Forest, LightGBM. The gradient-boosted model is tuned with Optuna (50 trials) against a deliberately business-shaped objective: **maximise recall subject to a precision floor** of 0.55. Maximising recall alone is degenerate (flag everyone, recall = 1.0), so the floor pins the model to an operating point a retention team can actually staff.

Model	Thr.	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.430	0.557	0.904	0.689	0.905	0.761
LightGBM	0.360	0.547	0.928	0.689	0.916	0.803
Random Forest	0.375	0.556	0.896	0.686	0.898	0.752
Decision Tree	0.444	0.561	0.858	0.679	0.879	0.700

Table 2. Held-out test performance at each model's recall-first operating threshold (archetype label included as a feature).

The standout figure is **recall of 0.928** for the boosted model — they catch roughly 93 of every 100 true churners. Precision sits near the 0.55 floor by design: the retention team accepts contacting some customers who would have stayed in exchange for missing very few who would have left. For context, the confusion matrix for the finalised LightGBM model at its 0.40 operating threshold flags **606 of 1,409** test customers as at-risk and misses only 27 of 374 real churners.

Finalised LightGBM @ 0.40	Predicted stay	Predicted churn
Actually stayed	748 (true negative)	287 (false positive)
Actually churned	27 (missed churner)	347 (caught churner)

Table 3. Confusion matrix for the production model. Only 27 churners missed against 347 caught — recall 0.91 at this operating point.

3.4 Explaining the risk with SHAP

A risk score alone cannot drive an intervention. The pipeline uses **SHAP** (SHapley Additive exPlanations) on the finalised LightGBM model to decompose each customer's churn probability into signed, per-feature contributions measured in churn log-odds. A positive SHAP value means the feature pushed this customer *toward* churning; a negative value means it anchored them. SHAP is additive, so one-hot categorical pieces are summed back to their original feature — the team reads "Contract," not "Contract_Month-to-Month" — and the archetype one-hot columns are removed after the fact so the drivers reported are genuine customer attributes, not the cluster label.

Across the churner population, the dominant drivers are consistent with domain intuition: a **month-to-month contract** is by far the strongest churn signal, followed by **lack of referrals**, **few or no dependents**, **high monthly charge**, and a high **monthly-to-total spend ratio** (a newer-customer signal). The example below is one predicted churner the model scored at 0.99.

Driver	This customer	SHAP	Effect
Contract	Month-to-Month	+1.456	raised risk
Latitude / location	32.857	+0.863	raised risk
Age	74	+0.593	raised risk
Number of Referrals	0	+0.502	raised risk

Driver	This customer	SHAP	Effect
Number of Dependents	0	+0.302	raised risk
Monthly Charge	\$95.45	+0.286	raised risk

Table 4. Top signed SHAP drivers for one predicted churner (probability 0.99, Archetype 3). Each value is that feature's push on churn log-odds for this specific customer.

3.5 From drivers to action — gpt-oss-120b via the Groq API

The final stage converts SHAP evidence into language a retention agent can use. For every predicted churner, the pipeline assembles a structured brief — the signed SHAP drivers, each driver's plain-English meaning, and the customer's archetype peer context — and sends it to a large language model. The model used is **gpt-oss-120b** (a 120-billion-parameter open-weights model), served through the **Groq API** for low-latency inference and orchestrated with **LangChain**-style prompt structuring (a fixed retention-analyst system prompt plus the per-customer brief as the user message). The LLM is instructed to treat SHAP as association, not proven cause, and to return exactly two sections.

Output	Granularity	What the LLM produces
Customer brief	One per at-risk customer	"Why they are at risk" (2–4 bullets tied to named SHAP drivers) + "Actions to prevent churn" (2–4 concrete offers/outreach moves, each addressing a driver, framed by archetype).
Archetype playbook	One per segment	A reusable retention playbook: segment profile + 3–5 prioritised plays, each with a trigger, an offer/action, and its business risk — built from the segment's mean SHAP drivers.

Table 5. The two LLM outputs. Briefs drive one-to-one outreach; playbooks let a team run a whole segment consistently.

For the 0.99-probability customer above, the model returned a brief that named the month-to-month contract, the high \$95.45 bill, the absence of referrals and dependents, and the customer's age (74) — then recommended a discounted annual contract with a no-penalty exit clause, a one-time bill credit with a cheaper bundle, a referral incentive, and a personal outreach call from a senior-care specialist. Each recommendation is tied to a specific driver and framed by the Archetype 3 peer pattern. The whole test set — all 1,409 customers, with archetype, probability, predicted label, top-3 drivers, and the LLM suggestion — is exported to a single spreadsheet the retention team opens and works straight down.

SECTION 4

How Clustering Powers the Suggestions

This is the crux of the design, and it is worth restating precisely. Their value is in the **explanation and action layer**, and it shows up in three concrete ways.

- **Peer context makes every score legible.** The LLM does not just see a customer's own drivers — it sees which archetype they belong to and how that segment behaves ("segment churn rate 54%, 920 customers"). A month-to-month contract means something different for a loyal Archetype-0 customer than for an isolated Archetype-3 one, and the advice is framed accordingly.
- **Segment-level playbooks turn one-off advice into scalable strategy.** By averaging SHAP drivers within each archetype, the pipeline produces one reusable playbook per segment. Instead of 606 disconnected suggestions, the retention team gets four coherent strategies — each with triggers, offers, and risks — that a campaign can be built around.

- **Archetypes prioritise where retention spend goes.** Churn rate ranges from 7.5% (Archetype 0) to 54.5% (Archetype 3). That spread tells an operator where a retention budget earns its highest return, and lets the same SHAP→LLM machinery be pointed first at the segments that are actually leaving.

SECTION 5

Outcome and Roadmap

The product delivers on its founding problem. A single classifier that missed too many churners has become a recall-first pipeline that catches roughly 93% of them, explains each risk in the customer's own terms, and drafts a specific retention move — all exportable to one spreadsheet a retention team can act on the same day. The honesty of the build (measured leakage removal, out-of-fold thresholds, stability-checked clusters) means the reported numbers are ones the operator can expect to reproduce in production rather than optimistic lab figures.

The clearest next step comes from the data audit itself: the strongest *stated* reason customers give for leaving — the attitude of a support agent (13% of churners) — is not measured anywhere in the current tables. Adding support-interaction data (ticket volumes, handle times, sentiment) is the single most promising way to lift recall further and give the SHAP→LLM layer an even sharper set of drivers to act on. Calibrated probabilities and an A/B test of the LLM-generated offers against a control would close the loop from prediction to measured revenue saved.

SECTION 6

References

Website Link: <https://telco-customer-churn.netlify.app/>